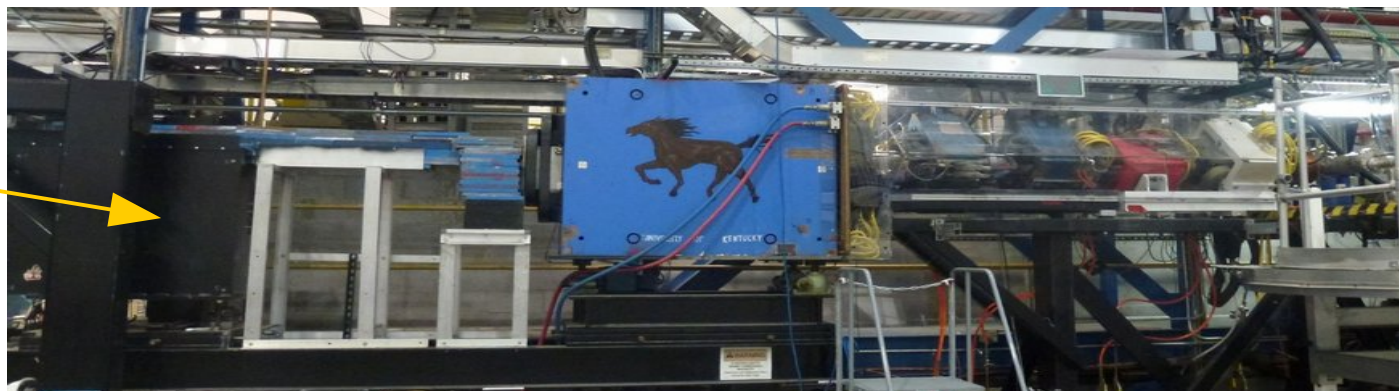
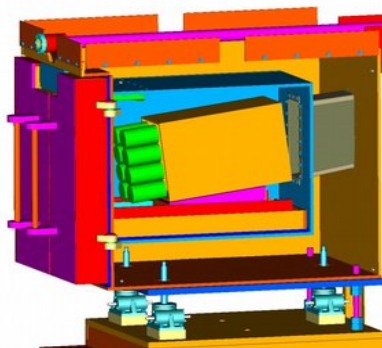


Möller Polarimeter DAQ Overview

Roman Pomatsalyuk, Olexandr Glamazdin

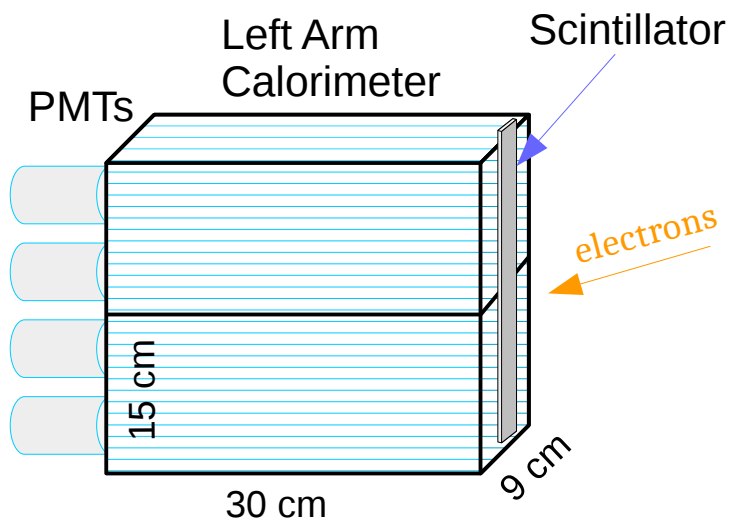
Hall A Jefferson Lab,
NSC “Kharkov Institute of Physics and Technology”

Møller Polarimeter Detector



The Møller polarimeter detector is located in the shielding box downstream of the dipole and consists of two identical parts placed symmetrically about a vertical plane containing the beam axis, thus enabling coincidence measurements.

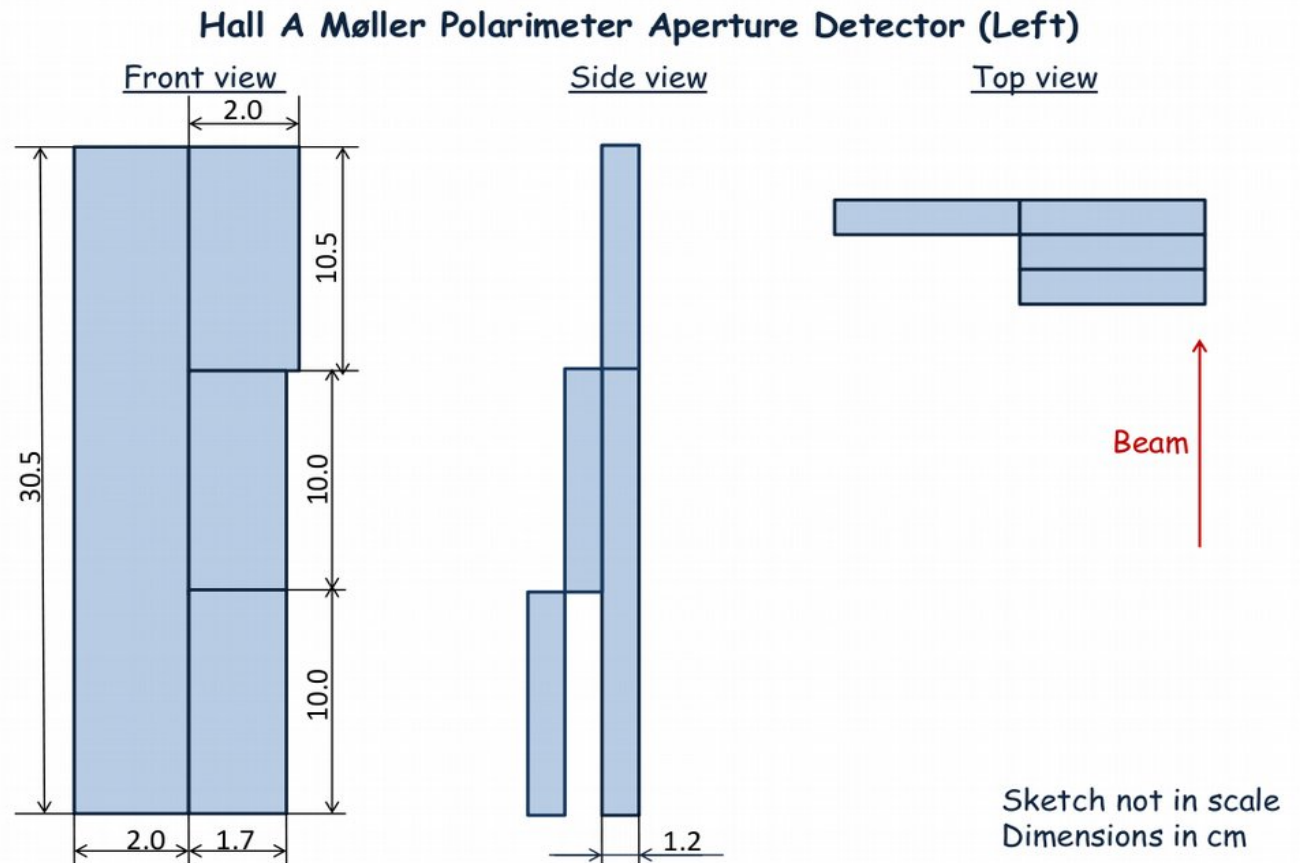
Each part of the detector includes:



- A “spaghetti” type (lead+scintillating fiber) calorimeter consisting of 2 blocks $9 \times 15 \times 30 \text{ cm}^3$, each separated into 2 channels equipped with photomultiplier tubes (Photonics XP2282B - 2 inch). Thus, the vertical aperture is segmented into 4 calorimeter channels.
- An aperture detector consists of four scintillators with light guides and Hamamatsu R4124 (13 mm diameter) photomultiplier tubes connected to each segment. Size of the aperture assembled detector is $31 \times 4 \times 3.6 \text{ cm}^3$.

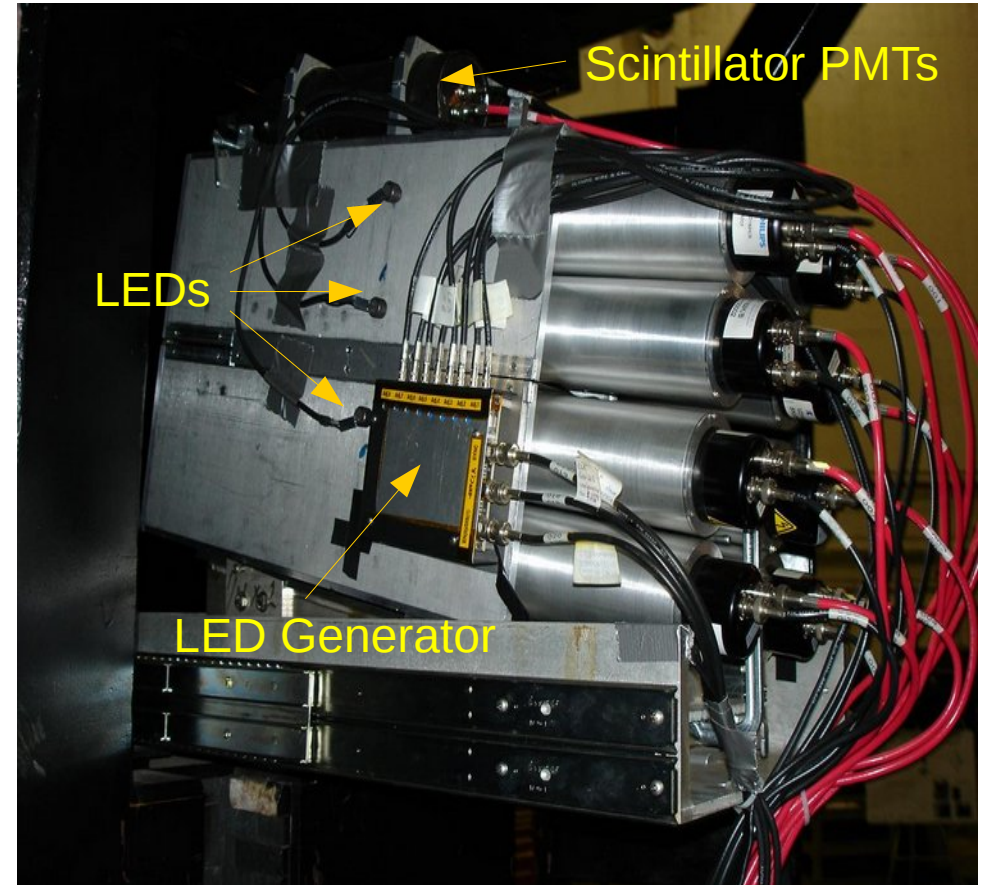
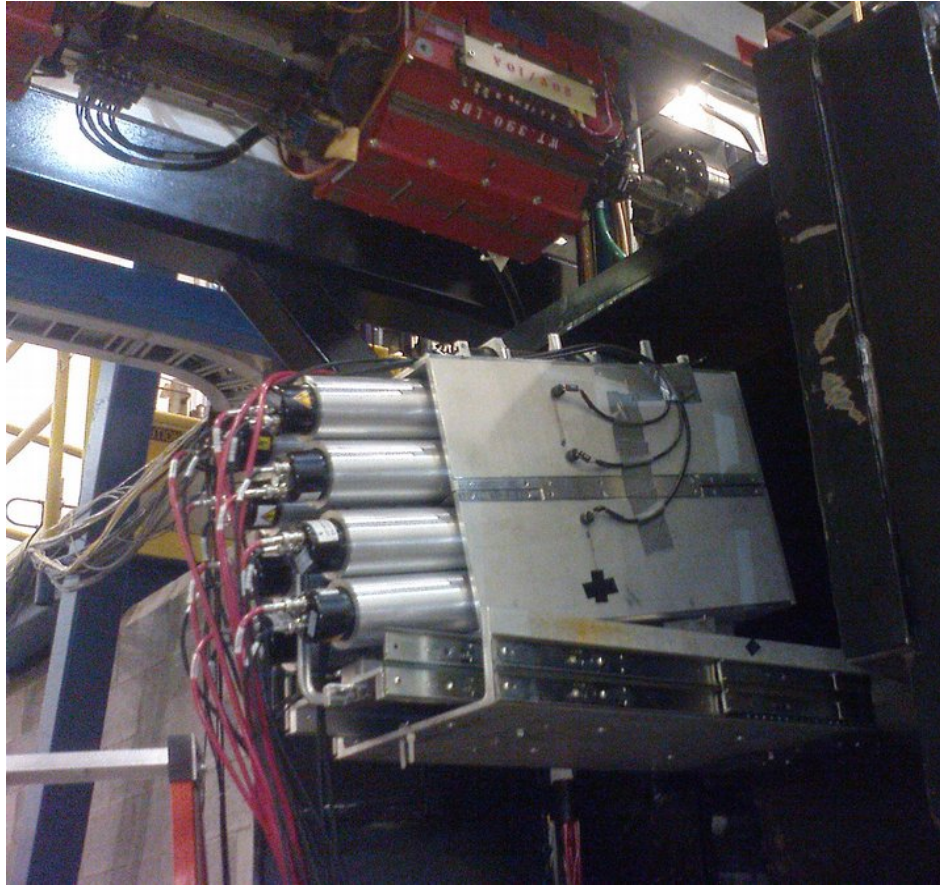
Total detector channels is 16: 8 calorimeter channels, 8 scintillator (aperture) channels.

Aperture Scintillators



- An aperture detector consists of four scintillators with light guides and Hamamatsu R4124 (13 mm diameter) photomultiplier tubes connected to each segment. Size of the aperture assembled detector is 31×4×3.6 cm³.

Detector Picture (out of shielding box)



Old DAQ Electronics

Two DAQ's: old DAQ and FADC based DAQ working in parallel (CODA 2.5)

Old DAQ electronics are located in several crates in the Hall A:

1. VME crate - for DAQ and slow control;
2. CAMAC crate - for the trigger and data handling;
3. 2 NIM crates - for the trigger and coincidence logic;
4. HV crate LeCroy 1450 – HV supply for detector PMT's (slot 5,6).

Triggers types for CODA Readout.

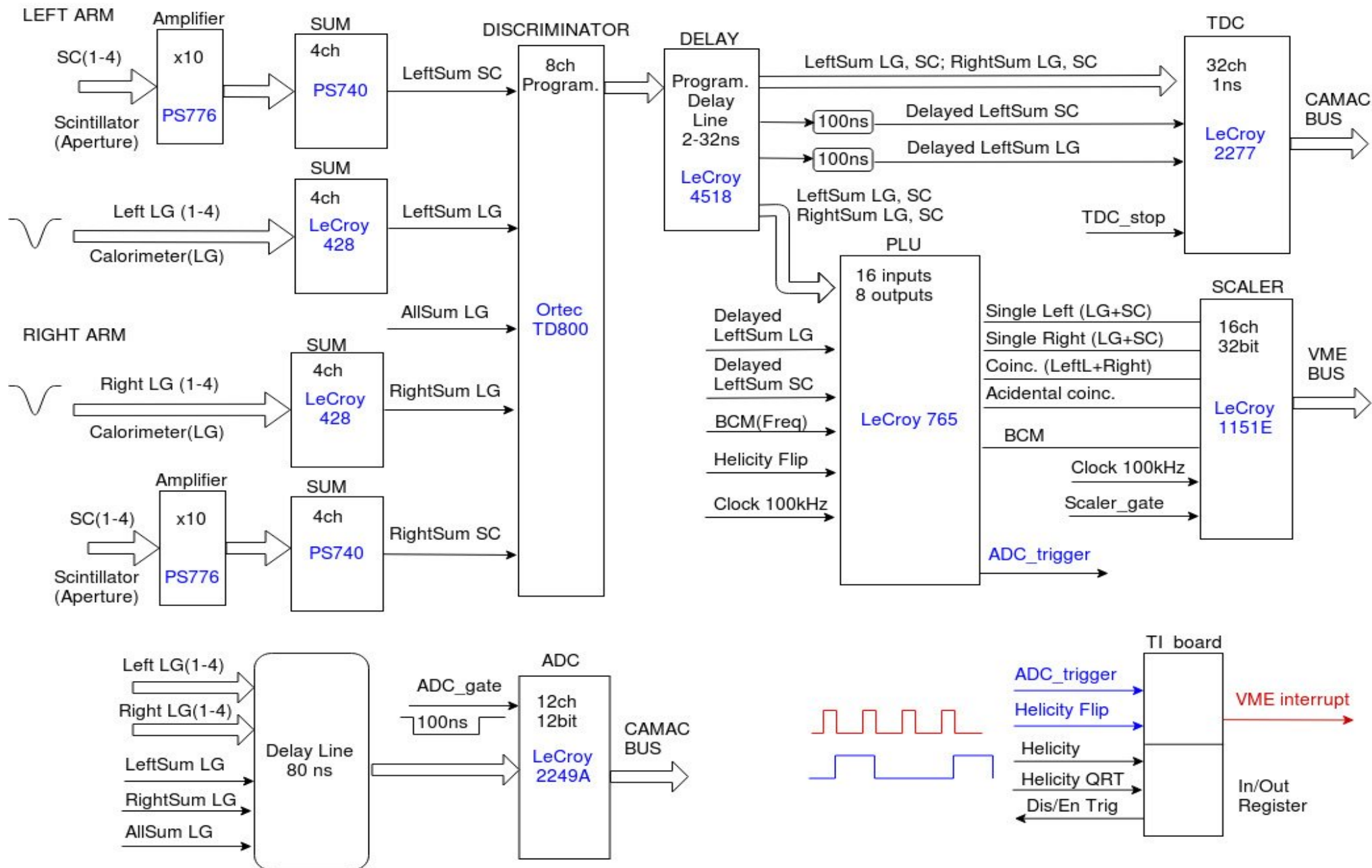
1. Helicity flipping trigger (30-1000Hz) - Scalers readout.
2. Event trigger (selected by PLU: Left/Right arm, Coincidence) - ADC, TDC readout.

Type 2 trigger is used while tuning up detector (high voltage and thresholds setup). Then it is suppressed.

The detector events rate for coincidence(left arm and right arm) is about 30-60kHz and for single arm is about 150-300kHz.

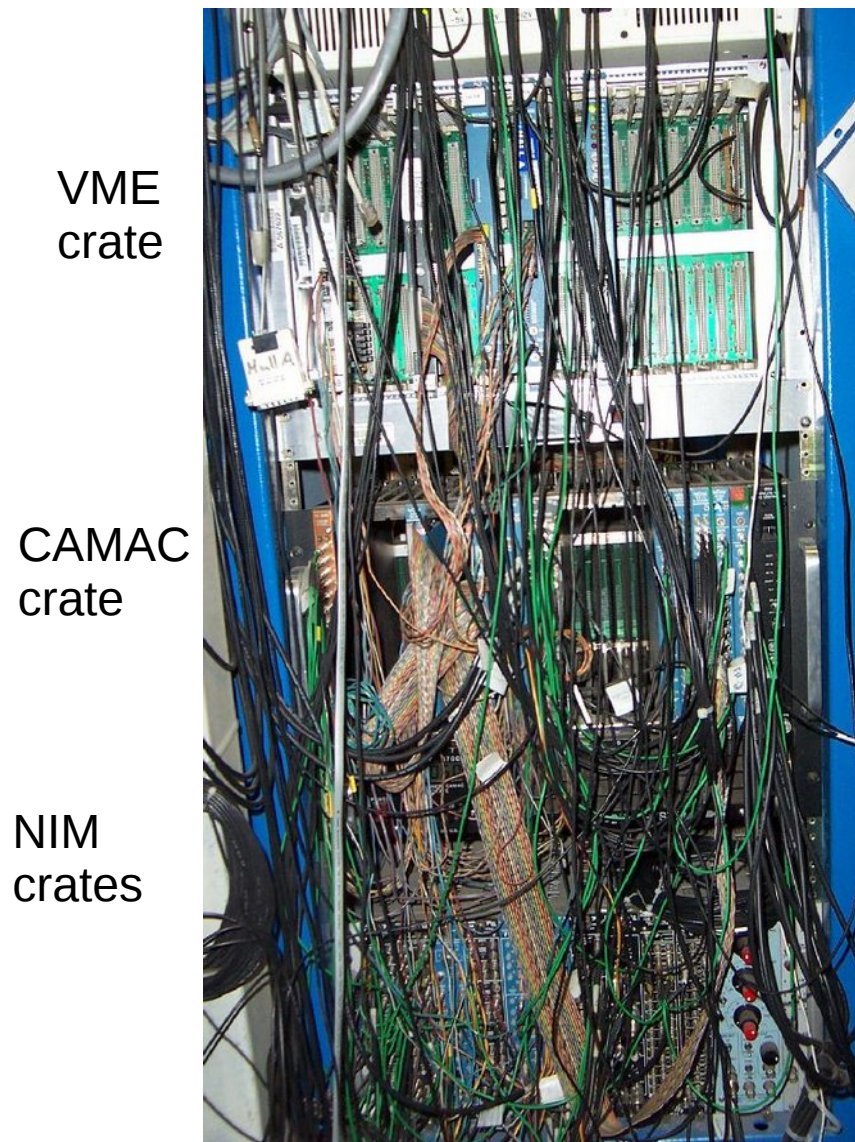
Old DAQ diagram

<http://hallaweb.jlab.org/equipment/moller/electronics.html>

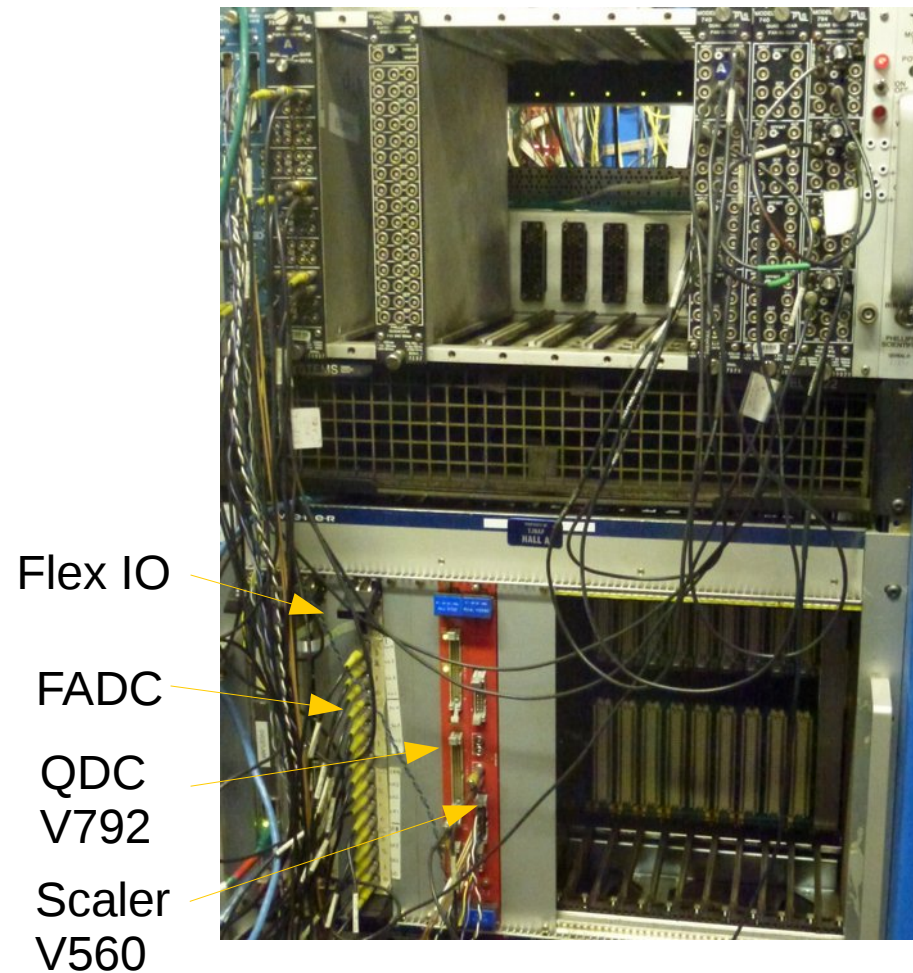


DAQs Picture

Old DAQ



FADC DAQ



Old DAQ Analyzing Software

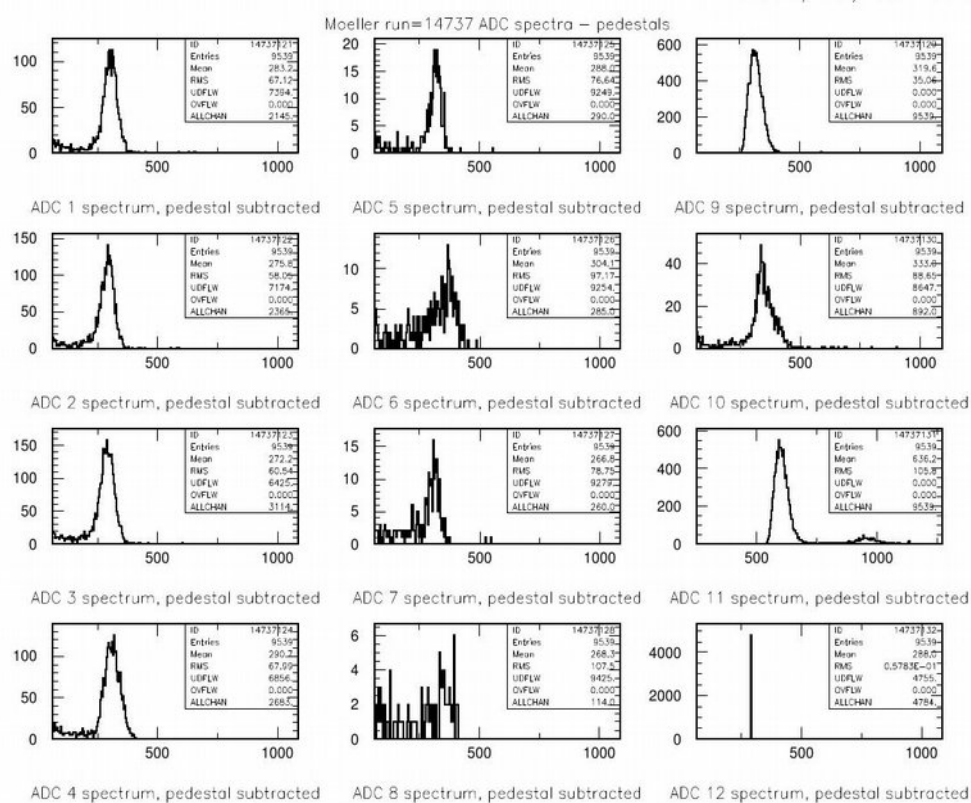
[https://hallaweb.jlab.org/wiki/index.php/Beam_Polarization_Measurement_\(How_to\)](https://hallaweb.jlab.org/wiki/index.php/Beam_Polarization_Measurement_(How_to))

- CODA and analyzer run on 'adaq1' computer.
- PAW based analyzer (Fortran) ~moller/paw/moller
- Few 'kumac' files with parameters to run
- Produce ntuples, plots and table with results of beam polarization measurements

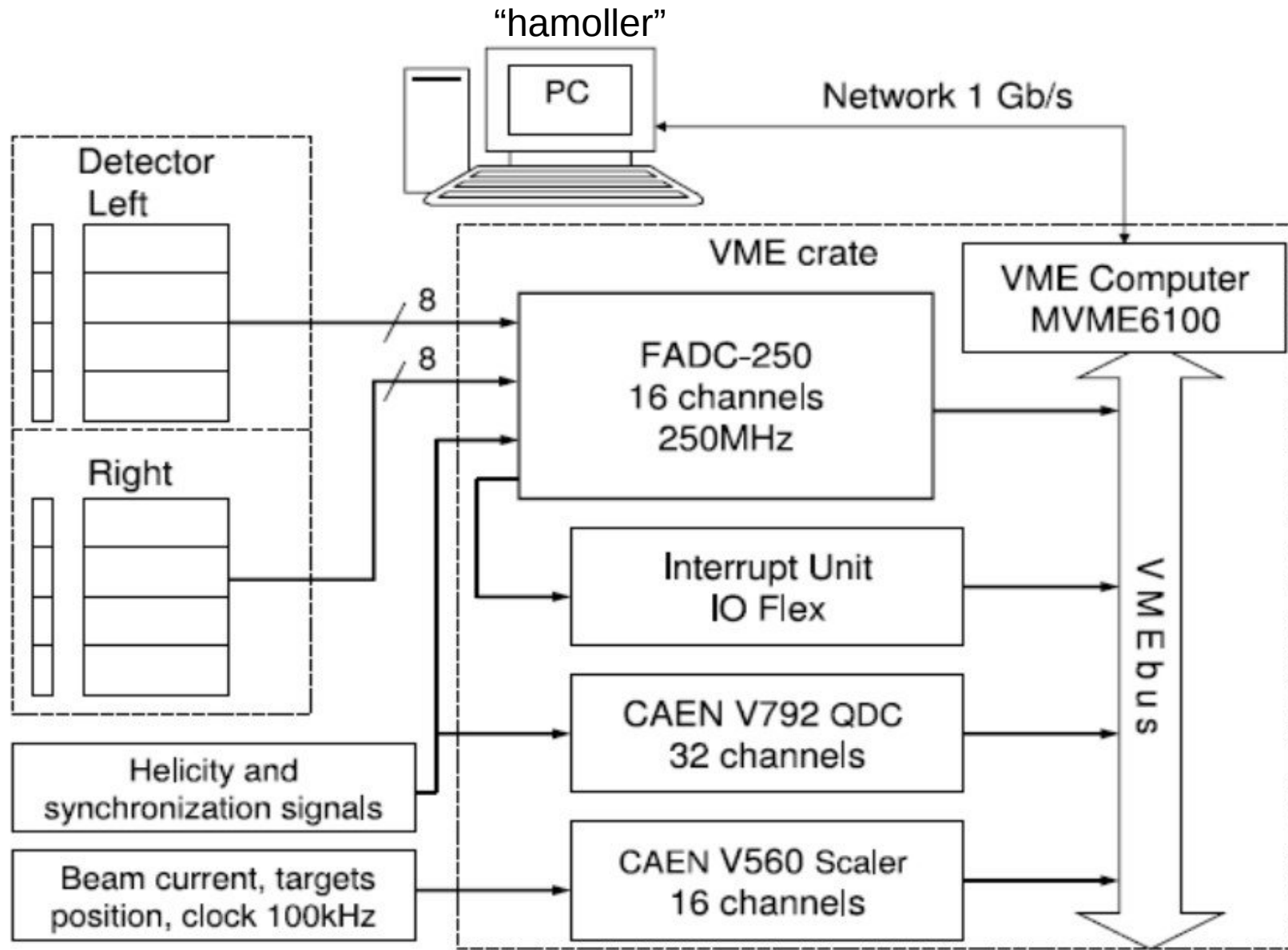
Moller measurements: rates/sec and asymmetries

run	Left	Right	Coinc.	Accid.	BCM	Clock	Cor.Asymm	Polarization	angl	An.Pow	P
* Target #3, 3T											
* Detector tune											
16010	238989	195901	59332	801.	34907.	101453	0.0542+/- 0.0005	0.8732+/- 0.0076	0.00		
16011	235980	195050	58493	809.	35008.	101463	0.0544+/- 0.0005	0.8764+/- 0.0083	0.00		
16012	235097	195284	58012	811.	34789.	101461	0.0536+/- 0.0007	0.8634+/- 0.0120	0.00		
16013	236830	194493	57399	819.	34804.	101464	0.0538+/- 0.0004	0.8662+/- 0.0067	0.00		
16014	216226	184401	52693	710.	34701.	101464	0.0528+/- 0.0005	0.8495+/- 0.0086	0.00		
* Q3=105.75A											
16015	216965	184341	52845	718.	34831.	101410	0.0535+/- 0.0005	0.8611+/- 0.0086	0.00		
16016	213822	188094	53738	716.	34846.	101460	0.0537+/- 0.0004	0.8653+/- 0.0060	0.00		
16017	207065	193588	54672	717.	34650.	101459	0.0540+/- 0.0004	0.8703+/- 0.0060	0.00		
16018	208253	193626	54738	718.	34732.	101436	0.0543+/- 0.0004	0.8736+/- 0.0060	0.00		
16019	209201	194083	54918	731.	34885.	101461	0.0538+/- 0.0004	0.8655+/- 0.0060	0.00		
16020	209595	194203	55142	730.	34926.	101458	0.0541+/- 0.0004	0.8703+/- 0.0060	0.00		
16021	208409	194322	54958	723.	34856.	101459	0.0537+/- 0.0004	0.8653+/- 0.0060	0.00		
* Dead time meas.											
16022	233243	212731	4488	16.	34964.	101456	-0.0006+/- 0.0015	-0.0089+/- 0.0238	0.00		
16023	234587	211587	4498	16.	34972.	101457	-0.0005+/- 0.0020	-0.0083+/- 0.0327	0.00		
16024	225231	204058	4507	16.	34854.	101463	-0.0005+/- 0.0017	-0.0085+/- 0.0277	0.00		
* Q3=58.2A											
16025	186050	182404	45747	611.	34864.	101458	0.0548+/- 0.0002	0.8817+/- 0.0038	0.00		
16026	186861	180975	45535	607.	34808.	101442	0.0545+/- 0.0003	0.8769+/- 0.0046	0.00		
16027	187645	181126	45568	609.	34877.	101457	0.0547+/- 0.0003	0.8802+/- 0.0042	0.00		
16028	188052	180475	45457	608.	34873.	101459	0.0547+/- 0.0003	0.8811+/- 0.0044	0.00		
* Q3=20.6A											
16029	174267	169354	25489	526.	34847.	101458	0.0542+/- 0.0004	0.8701+/- 0.0062	0.00		
16030	197863	191675	28961	672.	39907.	101463	0.0542+/- 0.0004	0.8705+/- 0.0060	0.00		
16031	202170	195573	29555	697.	40788.	101460	0.0537+/- 0.0007	0.8614+/- 0.0116	0.00		
16032	211489	216598	31072	809.	43604.	101461	0.0541+/- 0.0004	0.8690+/- 0.0062	0.00		
16033	211300	215166	30854	804.	43403.	101461	0.0546+/- 0.0004	0.8770+/- 0.0061	0.00		
* Dead time meas.											
16034	144267	133398	4497	13.	23247.	101462	-0.0006+/- 0.0021	-0.0103+/- 0.0332	0.00		
16035	264965	244367	4493	26.	43274.	101461	-0.0007+/- 0.0021	-0.0110+/- 0.0333	0.00		

2010/10/03 17.26



FADC based DAQ Diagram

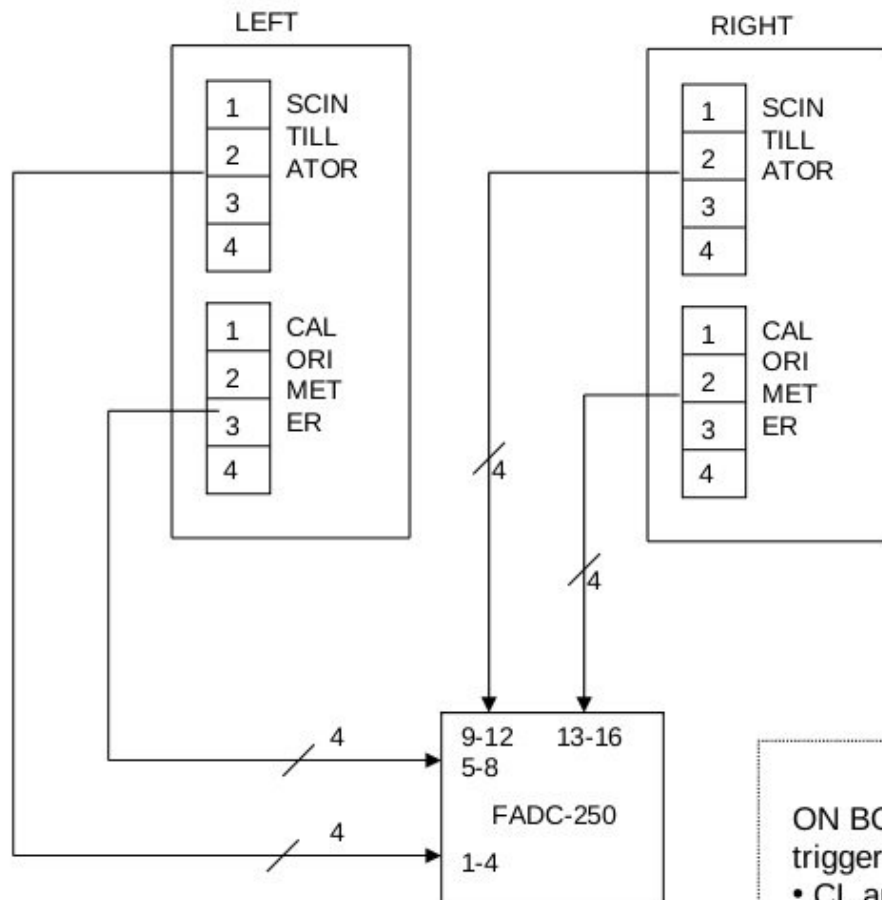


FADC module

The FADC implementation does the following:

- Make logical signals when combination of the input signals exceed certain thresholds;
- Write out the FADC “event” data using a logical signal for the trigger with an adjustable prescaler, at a rate up to the maximum rate allowed by the existing FADC systems (up to 160kHz);
- Makes internal scalers in the FADC unit counting the coincidence signal; the scalers are readout at every helicity cycle;
- The readout includes the “event triggers” (or “data_triggers” <160kHz) and the “helicity triggers” (30 – 2000Hz). Both additionally contain the helicity bit, the QRT bit, helicity window counts;
- The “helicity trigger” readout includes additional external scalers/QDC for the BCM, pulser, BPM, target encoders etc.

FADC Signal Processing Logic



CL – Sum of Calorimeter Left (1-4)
 CR – Sum of Calorimeter Right (1-4)
 SL – Sum of Scintillator Left (1-4)
 SR – Sum of Scintillator Right (1-4)

“Data trigger”

When TRIGGER condition is met, send data that cause TRIGGER.

TRIGGER condition (or):

CL AND CR prescaled from 1 to at least 1000

CL prescaled from 1 to at least 1000

CR prescaled from 1 to at least 1000

$$CR = \sum_{i=1,4} \sum_{j=1,2} P_i^j > \text{threshold}$$

$$CL = \sum_{i=1,4} \sum_{j=1,2} P_i^j > \text{threshold}$$

$$SL = (\sum_{j=1,2} S1^j > \text{threshold}) \text{ or } (\sum_{j=1,2} S2^j > \text{threshold}) \text{ or } (\sum_{j=1,2} S3^j > \text{threshold}) \text{ or } (\sum_{j=1,2} S4^j > \text{threshold})$$

$$SR = (\sum_{j=1,2} S5^j > \text{threshold}) \text{ or } (\sum_{j=1,2} S6^j > \text{threshold}) \text{ or } (\sum_{j=1,2} S7^j > \text{threshold}) \text{ or } (\sum_{j=1,2} S8^j > \text{threshold})$$

“Helicity trigger”

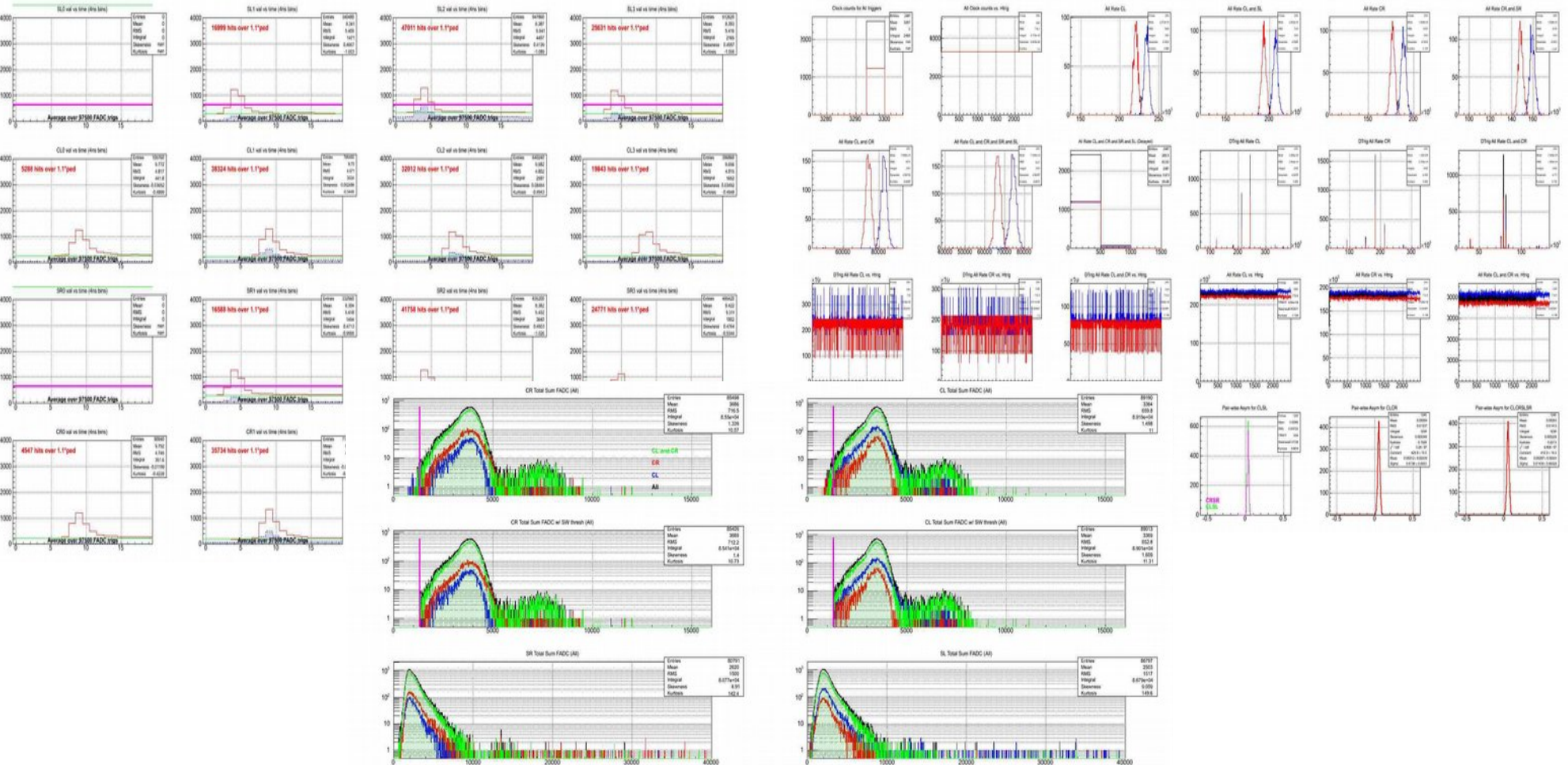
ON BOARD SCALER (COUNTER) to be read out by a separate trigger (helicity and gate bits) at the helicity cycle of 30 to 2kHz.

- CL and CR
- CL and SL
- CR and SR
- CL and CR and SL and SR
- ~~• CL and CR and (SL and SR) delayed > 100 ns~~
- (CL and CR and SL and SR) delayed > 100ns
- CL
- CR

FADC Software

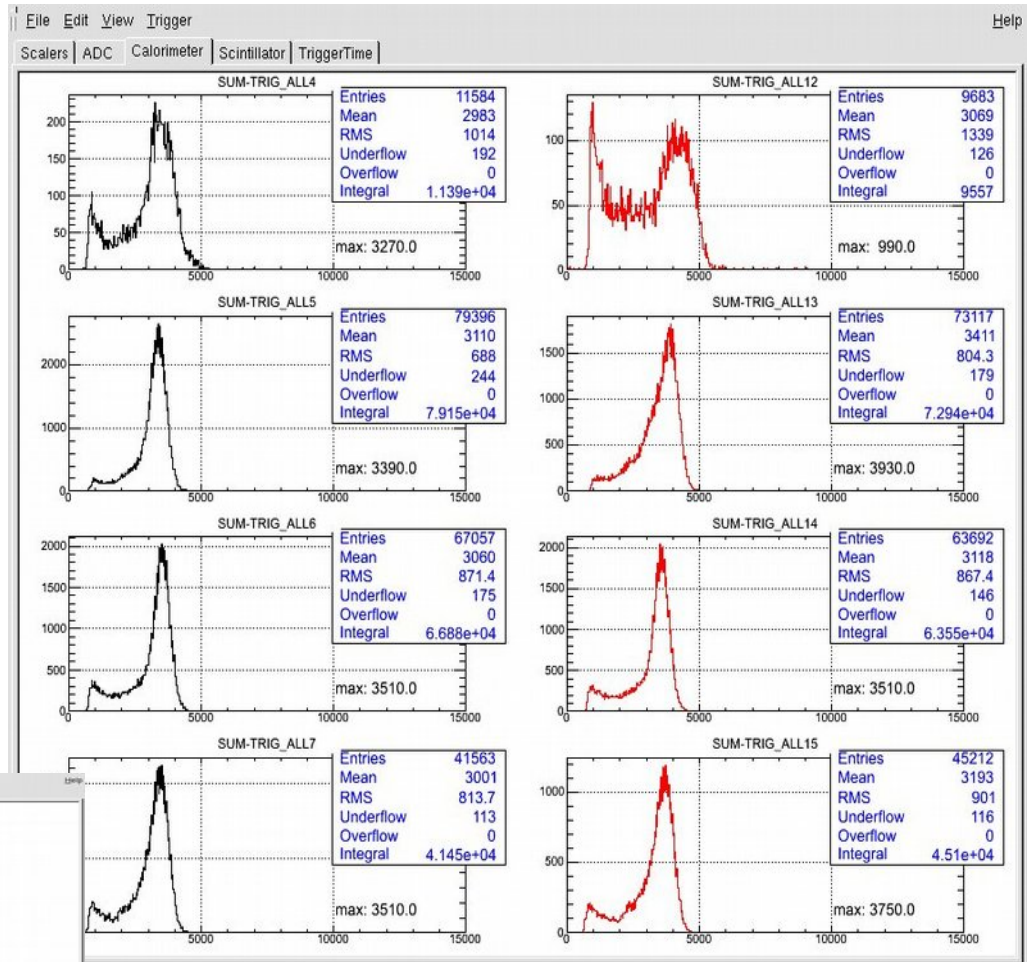
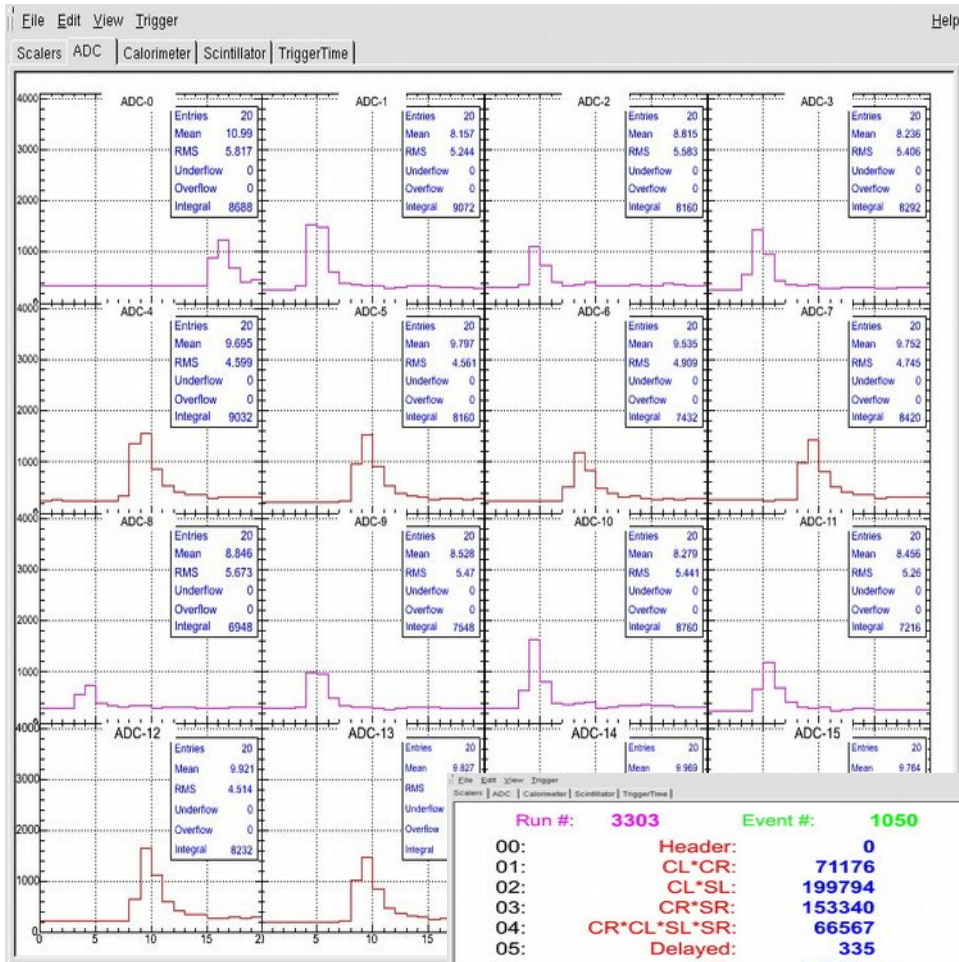
Docs: https://hallaweb.jlab.org/wiki/index.php/FADC_DAQ

- CODA and analysis run on 'hamoller' computer under "moller" account
- Podd/ROOT based analyzer
- ROOT macros to produce table and plots with results of measurements



Online FADC Monitor

Check signals from FADC during run



FADC V2 Status

Docs: https://hallaweb.jlab.org/wiki/index.php/FADC_DAQ_V2_2018

- Firmware and library developed by JLab electronics/CODA group in 2017-2018 (Hai Dong, Ed Jastrzembski, Bryan Moffit)
- Add PLU unit with 10 inputs, 8 outputs with scalers (8-channels). The scalers are readout at every “helicity trigger”.
- Test setup is in Test Lab SBS DAQ platform (Feb 2018, Linux VME CPU).
 - Input signals produced from internal FADC generator(“PlayBack mode”) or from external generator, helicity trigger – internal 30Hz.
- First version of new FADC module has some issues with processed logic and firmware/library under upgrade now.
- Current software (FADC has different setup/readout interface from old one):
 - standalone test programs
 - CODA readout list
- Work in progress on replay software to produce 'root' files.

To Do List

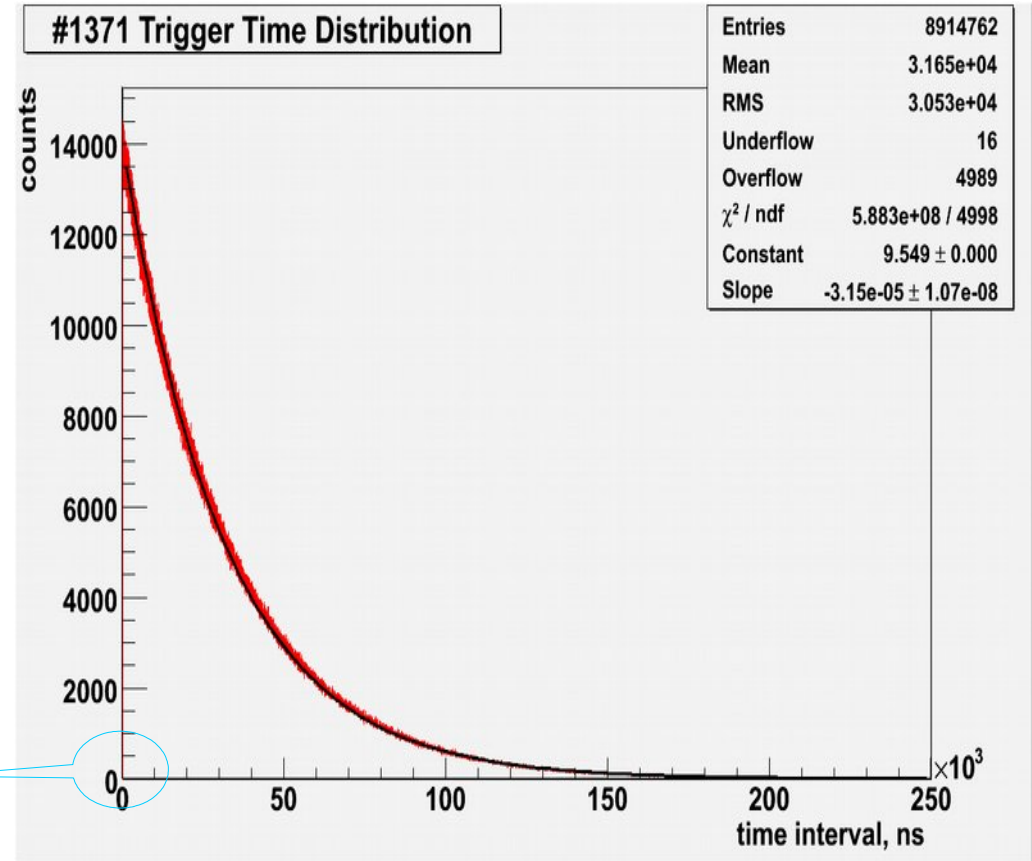
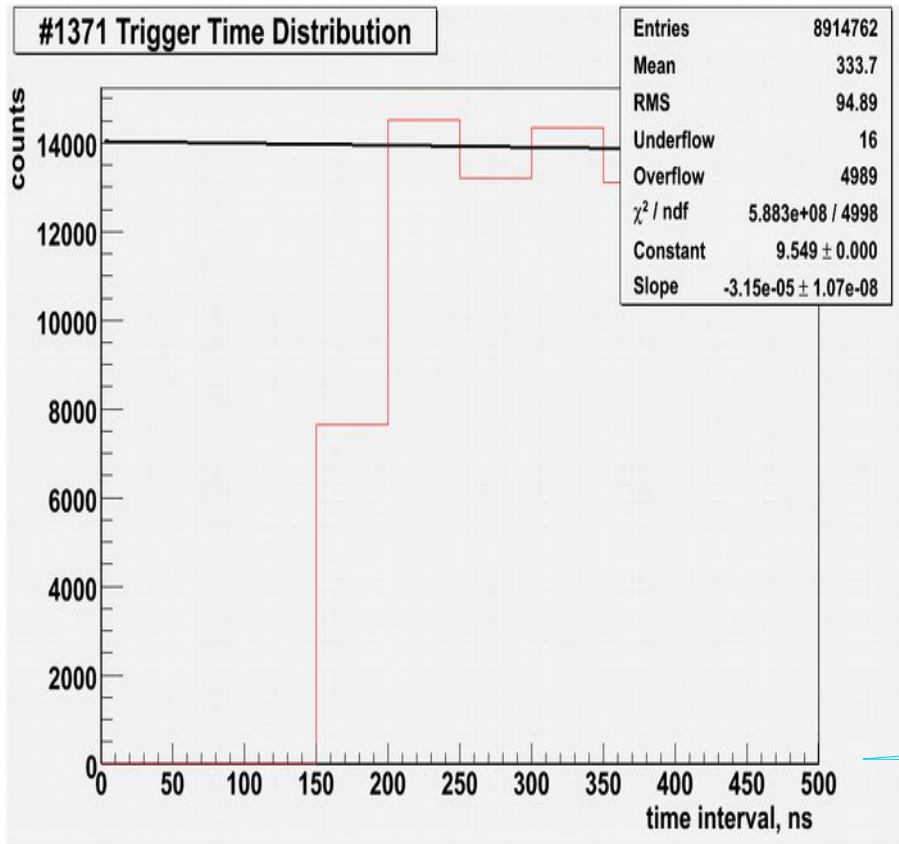
- Test new FADC with random generator in Test Lab
- Deploy to Hall A environments (VxWorks VME CPU, external signals, CODA configuration)
- Development/Upgrade of
 - CODA readout list for DAQ (FADC, QDC, Scaler)
 - Podd/ROOT based analyzer
- Testing with real signals, settings optimization
- Documentation
- 'hamoller' computer – upgrade/replace (disks issues)

Thank you for your attention!

Thank you for your attention!

FADC data trigger time distribution

There is a free running timer (48bit, 250MHz) on-board of the FADC. The time stamp of the every event (trigger) is recorded within of event data block.



Right: The time intervals distribution between of FADC events.

Left: Scaled time interval 0-500ns. Time shift about 150ns from zero time, that corresponds to the dead-time of the FADC.